

Juan David Vela Coronado

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Backend-focused Systems Engineering student specializing in building scalable, server-side systems and APIs for **AI-powered products**. Hands-on experience designing **microservices architectures** and **LLM/agent inference pipelines** with **Python (Flask)**, **Node.js (NestJS)**, and **Java (Spring Boot)**, integrated with **AWS** cloud services.

PROJECT EXPERIENCE

AIMO — Generative AI Conversational Assistant

2026

Backend Developer

Live Deployment

- Built the backend for a **Generative AI** assistant delivering real-time emotional support to university students, serving empathetic, safe, and contextualized responses based on the user's risk level.
- Designed a **3-agent cascading pipeline with continuous real-time evaluation** using **Python 3.13** and **Flask 3.0**, orchestrating inference across the **Groq API** and **AWS Bedrock** (Amazon Nova Pro).
- Engineered prompt orchestration and safety guardrails to guarantee reliable, context-aware model serving; **containerized with Docker** and deployed to production with a React frontend.

Community Action Boards (JAC) Management Platform

Popayán, Colombia

Tech Lead & Backend Developer

2026 • *Live Deployment*

- **Led a 5-developer team** building a **microservices platform** for the Popayán local government to manage Community Action Boards (JACs) and Asocomunales.
- Architected backend microservices with **NestJS (Node.js)** covering government user management, organizational data, and legacy data migration.
- Implemented **authentication and authorization with Keycloak** and **Google OAuth (Google Login)** for secure, multi-role access to a production government system.
- Delivered a **data migration module** to streamline updating institutional records; **containerized services with Docker** and shipped with a **React 19** frontend.

Alzheimer's Stage Classification — Machine Learning Pipeline

2025

Machine Learning Engineer

Best Poster Award — Universidad del Cauca (2026)

- Built a **machine learning pipeline** to predict Alzheimer's severity levels from medical imaging using **Convolutional Neural Networks (CNNs)** and **transfer learning**.
- Awarded **Best Poster** at Universidad del Cauca's student project showcase for the project's technical depth and impact.

Academic Project Management System

Popayán, Colombia

Backend Developer

2025

- Developed a **microservices-based backend** to manage academic projects involving students, companies, and coordinators.
- Built scalable services with **Java and Spring Boot**, ensuring loose coupling and service independence.
- Optimized database queries and inter-service communication to support concurrent users.

EDUCATION

Universidad del Cauca

Popayán, Colombia

Bachelor's Degree in Systems Engineering

Expected Graduation: 2027

Relevant Coursework: Software Engineering, Distributed Systems, Databases, Artificial Intelligence, Operating Systems.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript/TypeScript, Go.
- **Backend:** Spring Boot, NestJS (Node.js), Flask, REST APIs, Microservices, WebSockets, gRPC.
- **AI / ML:** LLM & Agent Pipelines, Deep Learning (CNNs, Transfer Learning), AWS Bedrock, Groq API.
- **Cloud & Security:** AWS, Keycloak, OAuth2 / Google Login, Role-Based Access Control (RBAC).
- **Databases:** PostgreSQL, Oracle Database, MariaDB.
- **Frontend:** React, Angular, TypeScript, HTML, CSS.

- **Tools & Practices:** Git, Docker, Postman, Linux, Agile (Scrum).
- **Languages (Spoken):** Spanish (Native), English (Intermediate — technical proficiency).

CERTIFICATIONS

Getting Started with Deep Learning — NVIDIA	<i>2026</i>
Credential ID: M0S7oiZMQcO9R966P9O6-Q	
Full Stack Empresarial con Spring Boot y Angular — DevSenior	<i>2025</i>
Credential ID: e5f3a1e1-5caf-47cb-833e-a6a5b8f3629a	
Junior Cybersecurity Analyst Career Path — Cisco	<i>2025</i>
Credential ID: a917b782-b2b7-47d4-bf1c-5d370a6df267	